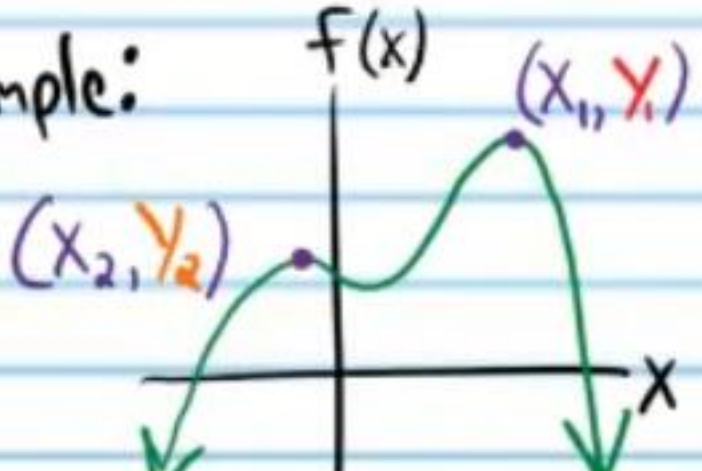


Applications of Differential Calculus (Maximum and Minimum Values)

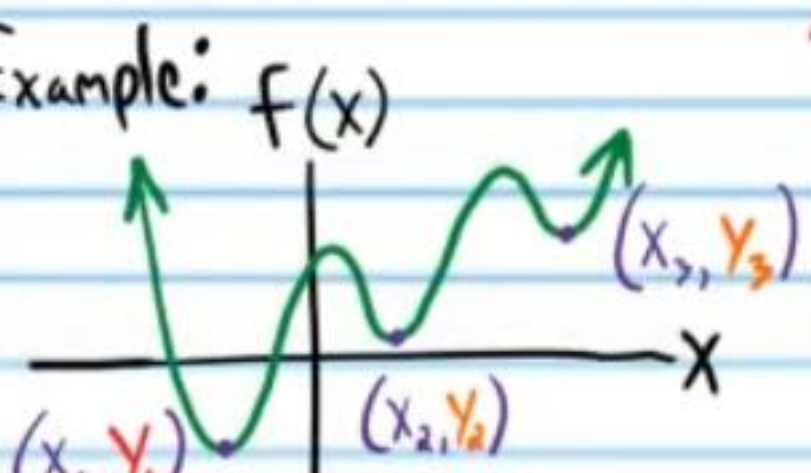
Absolute Maximum: An output of a function (or y-value of the graph of a function) that is greater than any other output of the function (or y-value of the graph of the function).

Local/Relative Maximum: An output of a function (or y-value of the graph of a function) that is greater than other outputs of the function that are near it but not necessarily greater than all outputs of the function. *Note:* Absolute maximums are also local maximums. *Note:* Mathematicians have decided that endpoints cannot be local maxima or minima. *Plural form of maximum and minimum.*

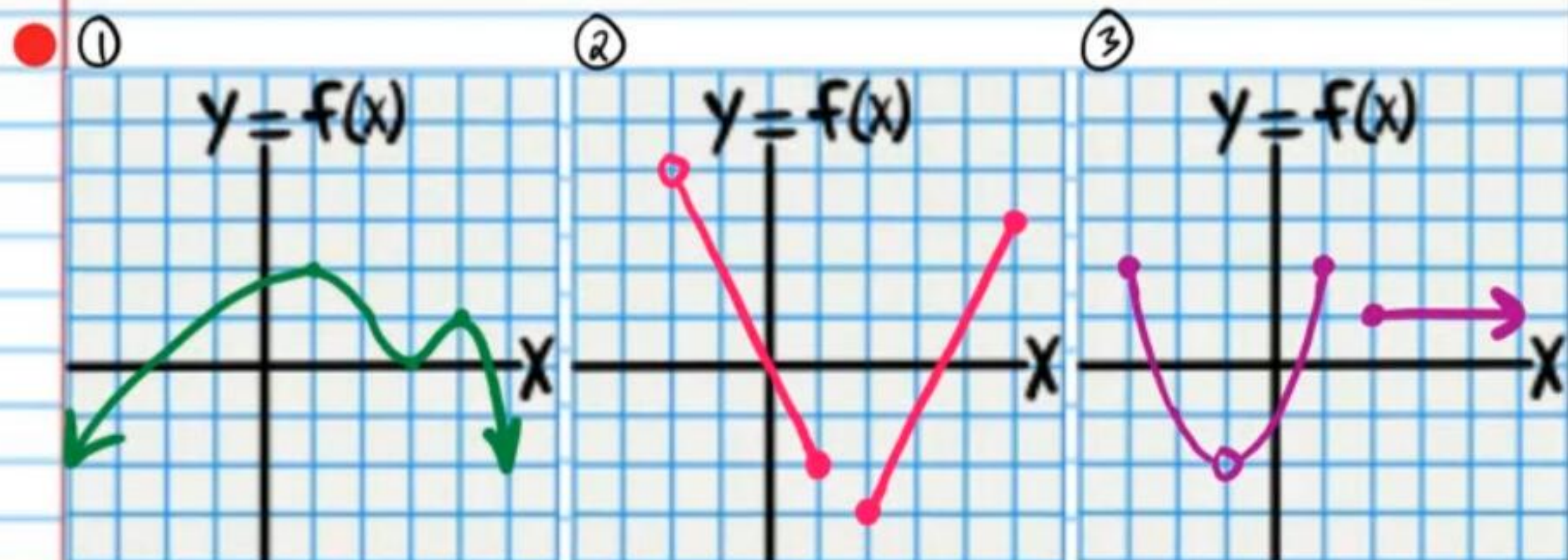
Example:  *"y₁" is the absolute maximum of the function (and a local maximum), assuming that the graph does not turn where we cannot see it.*
"y₂" is a local maximum of the function.

Absolute Minimum: An output of a function (or y-value of the graph of a function) that is less than any other output of the function (or y-value of the graph of the function).

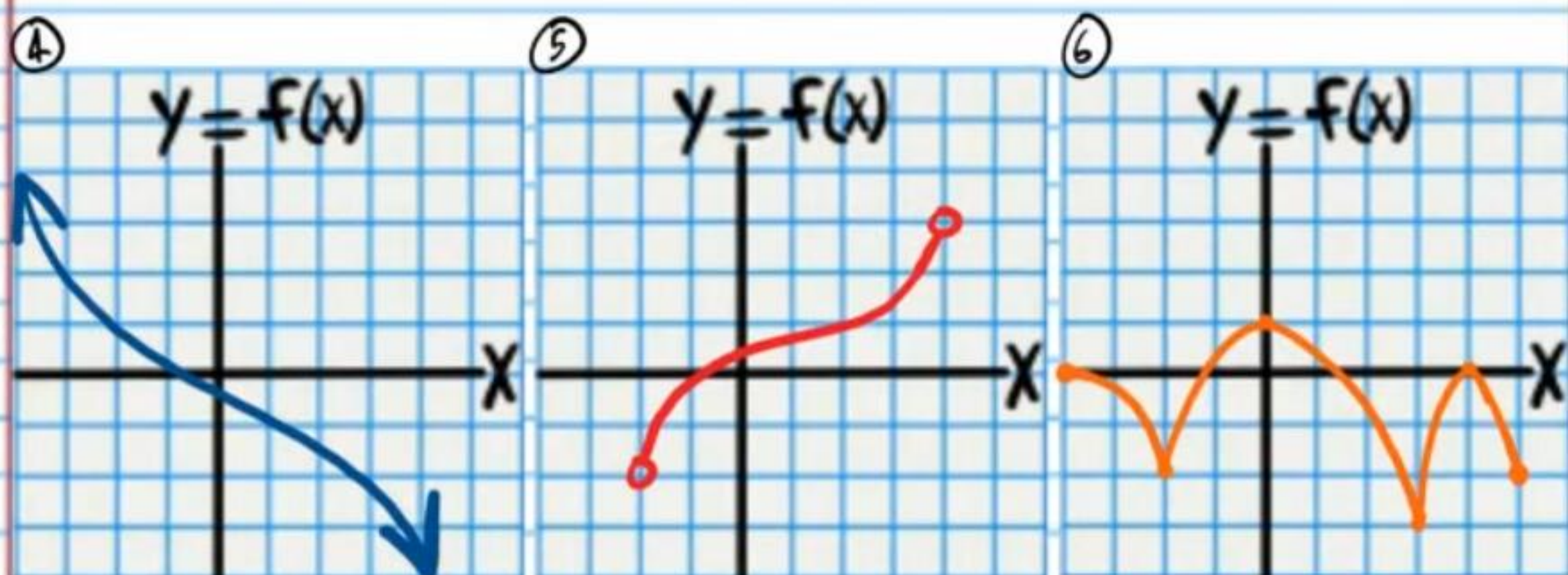
Local/Relative Minimum: An output of a function (or y-value of the graph of a function) that is less than other outputs of the function that are near it but not necessarily less than all outputs of the function. *Note:* Absolute minimums are also local minimums. *Note:* Mathematicians have decided that endpoints cannot be local maxima or minima. *Plural form of maximum and minimum.*

Example:  *"y₁" is the absolute minimum of the function (and a local minimum), assuming that the graph does not turn where we cannot see it.*
"y₂" and "y₃" are local minimums of the function.

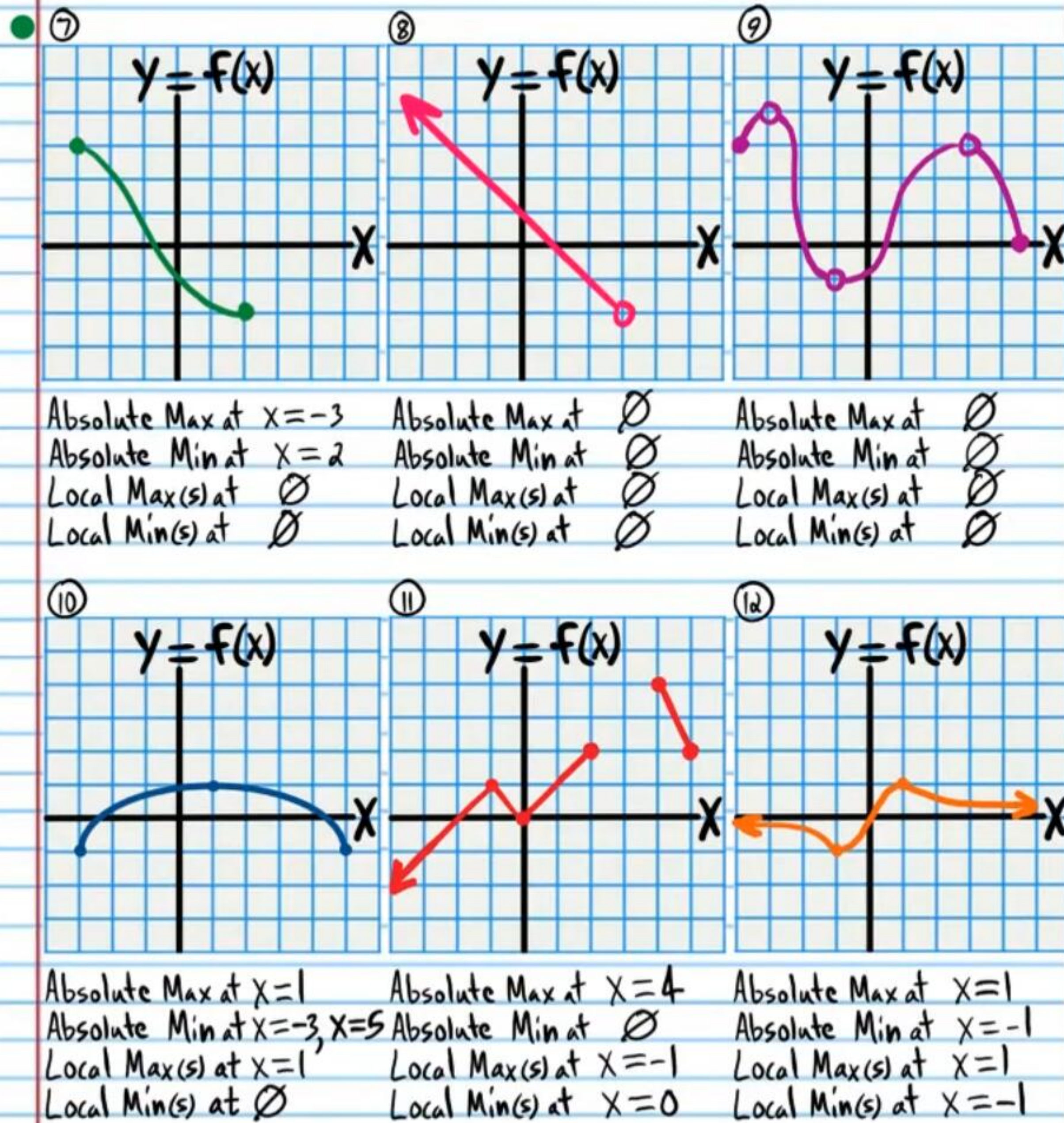
Write the x-value where each maximum or minimum occurs.
Assume that the direction of each graph continues as indicated by the arrows outside the window.



Absolute Max at $x=1$	Absolute Max at $\emptyset$	Absolute Max at $x=1, x=3$
Absolute Min at $\emptyset$	Absolute Min at $x=2$	Absolute Min at $\emptyset$
Local Max(s) at $x=4, x=1$	Local Max(s) at $\emptyset$	Local Max(s) at $\emptyset$
Local Min(s) at $x=3$	Local Min(s) at $\emptyset$	Local Min(s) at $\emptyset$



Absolute Max at $\emptyset$	Absolute Max at $\emptyset$	Absolute Max at $x=0$
Absolute Min at $\emptyset$	Absolute Min at $\emptyset$	Absolute Min at $x=3$
Local Max(s) at $\emptyset$	Local Max(s) at $\emptyset$	Local Max(s) at $x=0, x=4$
Local Min(s) at $\emptyset$	Local Min(s) at $\emptyset$	Local Min(s) at $x=-2, x=3$



Finding Maxima and Minima with Calculus

Example 1) Find the **local** minimum and **local** maximum values of the function below.

$$f(x) = \frac{1}{18}x^3 - \frac{1}{12}x^2 - x + \frac{1}{3}$$

$$f'(x) = \frac{1}{18}(3x^{3-1}) - \frac{1}{12}(2x^{2-1}) - x^{1-1} + 0$$

$$f'(x) = \frac{1}{6}x^2 - \frac{1}{6}x - 1$$

$$0 = \frac{1}{6}x^2 - \frac{1}{6}x - 1$$

$$0 = x^2 - x - 6$$

$$0 = (x-3)(x+2)$$

$$\swarrow \quad \searrow$$

$$x = -2 \text{ or } x = 3$$

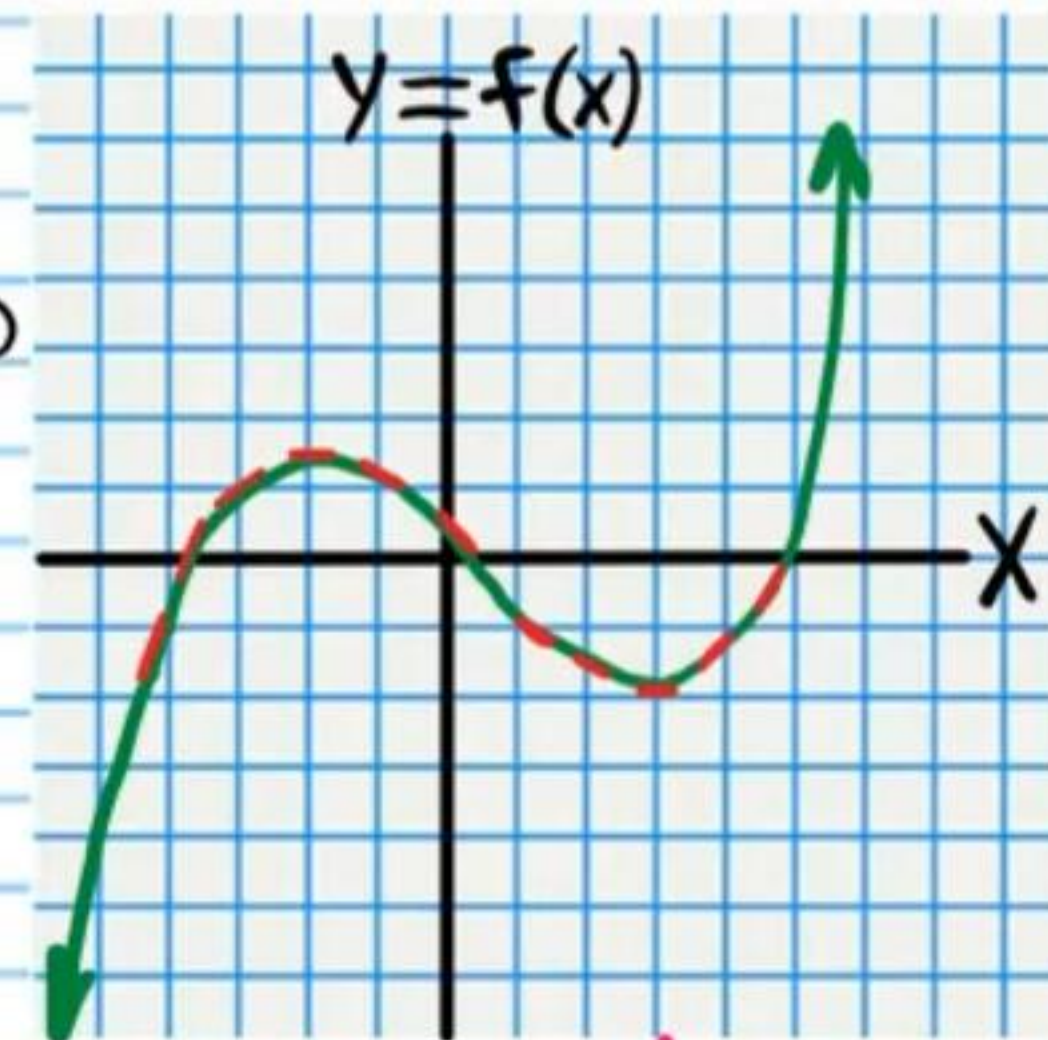
$$f(3) = \frac{1}{18}(3)^3 - \frac{1}{12}(3)^2 - 3 + \frac{1}{3} \quad f(-2) = \frac{1}{18}(-2)^3 - \frac{1}{12}(-2)^2 - (-2) + \frac{1}{3}$$

$$f(3) = \frac{3}{2} - \frac{3}{4} - 3 + \frac{1}{3}$$

$$f(3) = -1\frac{11}{12} \text{ Local Minimum}$$

$$f(-2) = -\frac{4}{9} - \frac{1}{3} + 2 + \frac{1}{3}$$

$$f(-2) = 1\frac{5}{9} \text{ Local Maximum}$$



Example 2) On the interval $[-5, 4]$, find the absolute minimum value and the absolute maximum value of the function from example 1.

$$f(x) = \frac{1}{18}x^3 - \frac{1}{12}x^2 - x + \frac{1}{3}$$

$$f(-5) = \frac{1}{18}(-5)^3 - \frac{1}{12}(-5)^2 - (-5) + \frac{1}{3}$$

$$f(-5) = \frac{1}{18}(-125) - \frac{1}{12}(25) + 5 + \frac{1}{3}$$

$$f(-5) = -\frac{125}{18} - \frac{25}{12} + 5 + \frac{1}{3}$$

$$f(-5) = -\frac{1500}{216} - \frac{450}{216} + \frac{15}{3} + \frac{1}{3}$$

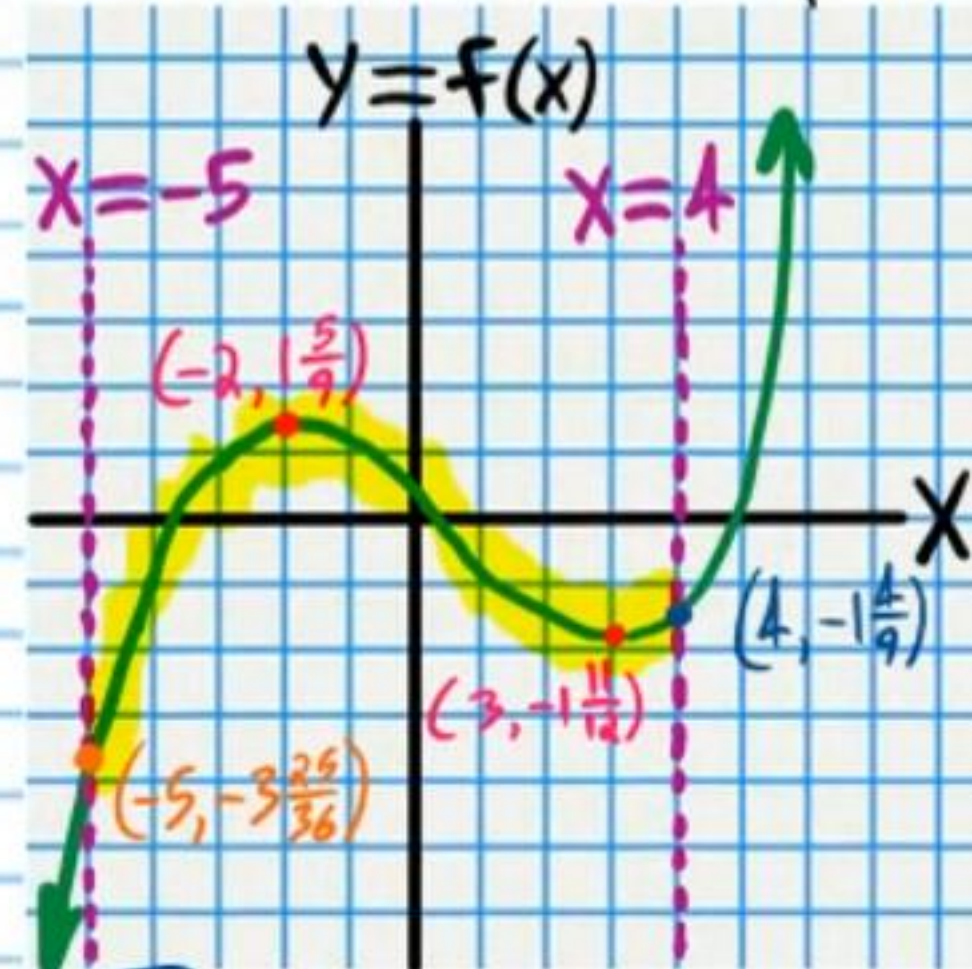
$$f(-5) = -\frac{1950}{216} + \frac{16}{3}$$

$$f(-5) = -\frac{798}{216}$$

$$f(-5) = -3\frac{25}{36}$$

$$f(4) = \frac{1}{18}(4)^3 - \frac{1}{12}(4)^2 - 4 + \frac{1}{3}$$

$$f(4) = \frac{1}{18}(64) - \frac{1}{12}(16) - 4 + \frac{1}{3}$$



$$f(4) = \frac{64}{18} - \frac{16}{12} - \frac{12}{3} + \frac{1}{3}$$

$$f(4) = \frac{32}{9} - \frac{4}{3} - \frac{12}{3} + \frac{1}{3}$$

$$f(4) = \frac{32}{9} - \frac{15}{3}$$

$$f(4) = -\frac{13}{9}$$

$$f(4) = -1\frac{4}{9}$$

Four Important Points on the Interval $[-5, 4]$.

$(-5, -3\frac{25}{36})$ $(-2, 1\frac{5}{9})$ $(3, -1\frac{11}{12})$ $(4, -1\frac{4}{9})$

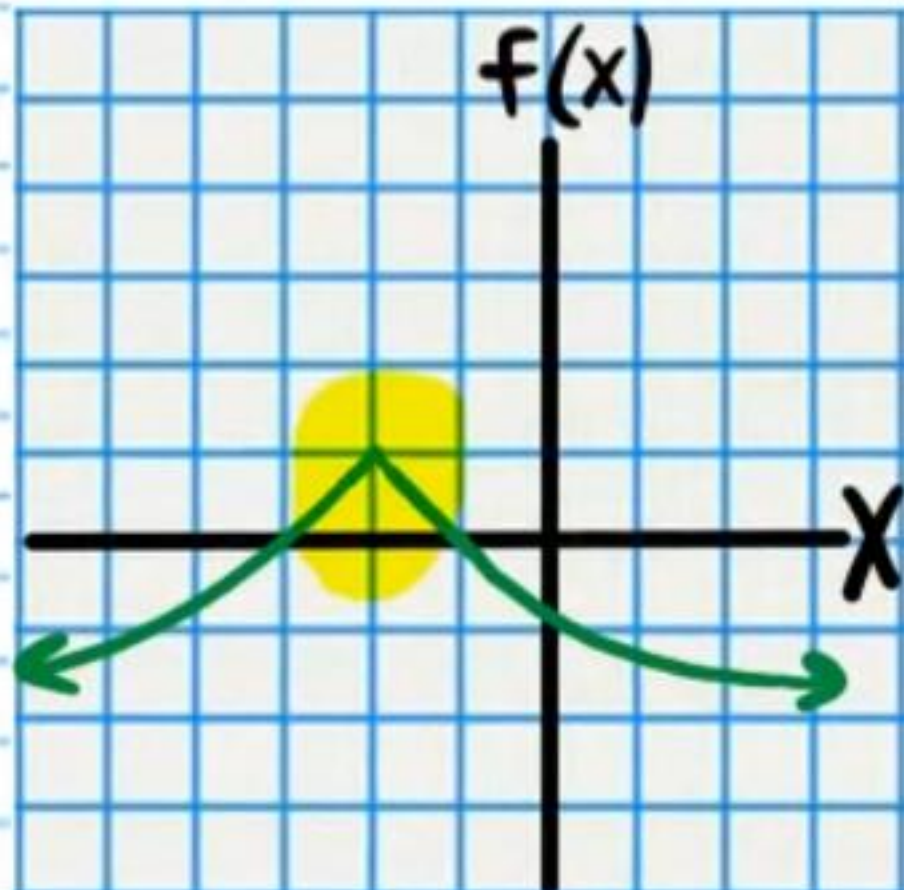
Absolute Max: $1\frac{5}{9}$ Absolute Min: $-3\frac{25}{36}$

Fermat's Theorem: If f has a local maximum or a local minimum at $x=a$, and $f'(a)$ exists, then $f'(a)=0$.

Example 3) Find the maximum value of the function below.

$$f(x) = 1 - (x+2)^{\frac{2}{3}}$$

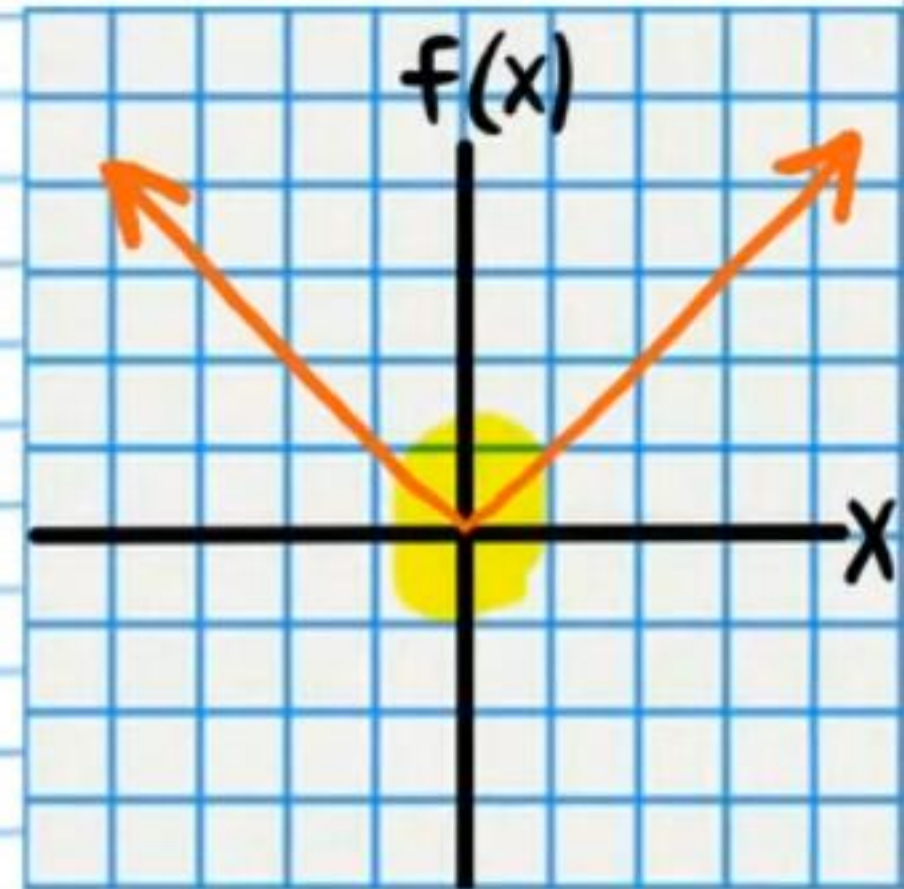
A maximum value of the function is located at $x=-2$, but $f'(-2) \neq 0$. Rather, $f'(-2)$ is undefined.



Example 4) Find the minimum value of the function below.

$$f(x) = |x|$$

A minimum value of the function is located at $x=0$, but $f'(0) \neq 0$. Rather, $f'(0)$ is undefined.



Critical Numbers: Inputs (i.e. x -values) where the derivative of a function is zero or undefined.

The locations of regions of interest of the graph of a function are described with x -values written in interval notation (e.g. (a, b) or $[a, b]$).

Critical Number Theorem: If a function has a maximum or minimum value, then the input (i.e. x-value) where the value occurs is a critical number. *This may not be true if the max/min is located at an endpoint of a finite interval.*

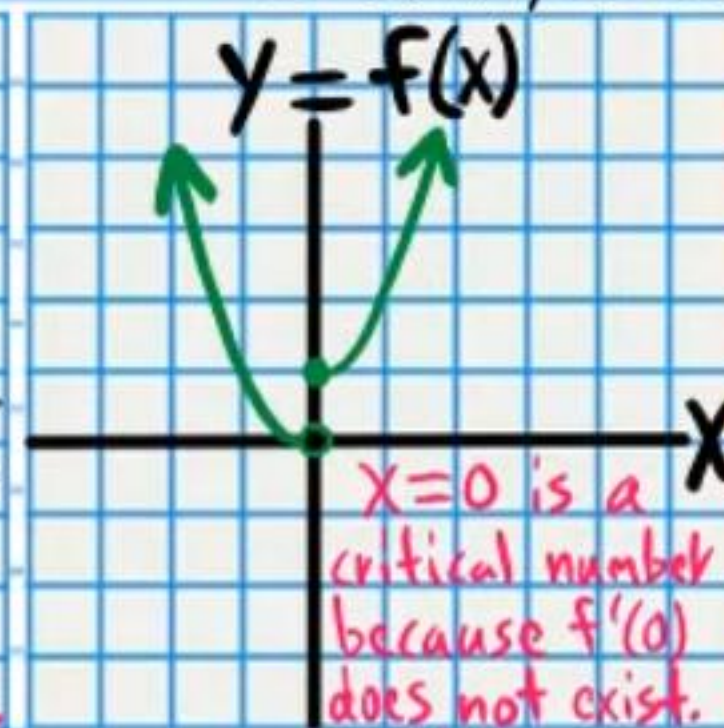
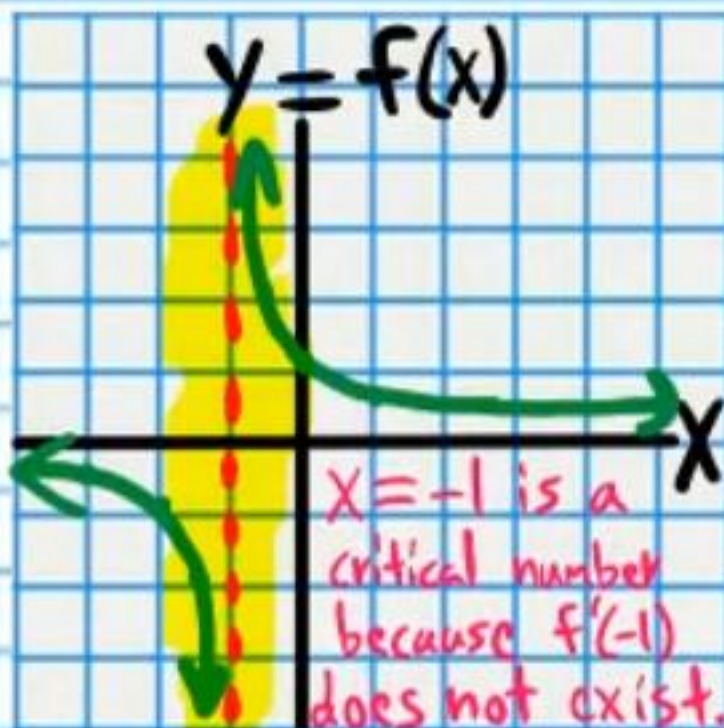
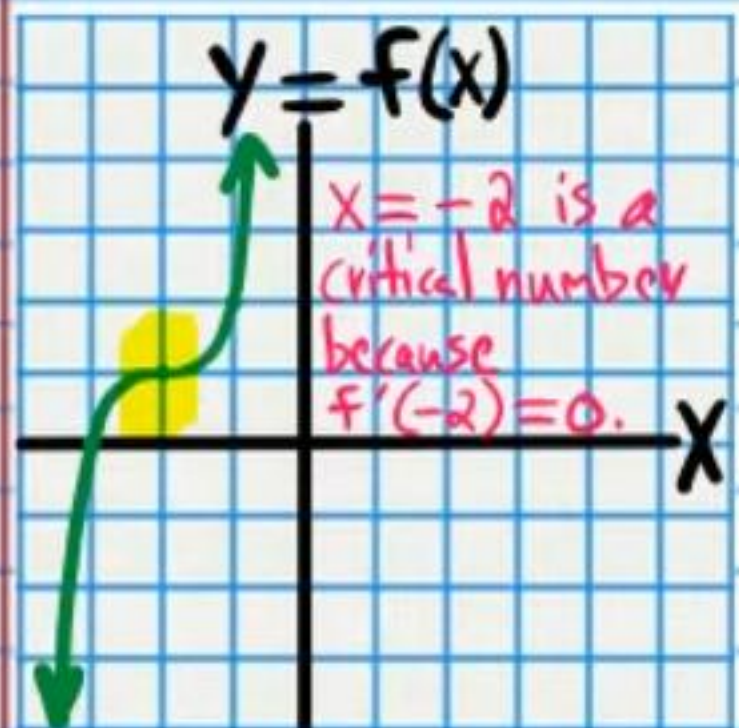
Warning
If the derivative of a function is zero or undefined at some x-value, this does not prove that a maximum or minimum value occurs at this x-value. It only suggests that a maximum or minimum may be present.

Examples where Critical Numbers Are Present but No Maximum or Minimum Value Occurs

$$f(x) = x^3 + 6x^2 + 12x + 9$$

$$f(x) = \frac{1}{x+1}$$

$$f(x) = \begin{cases} x^2, & x < 0 \\ x^2 + 1, & x \geq 0 \end{cases}$$



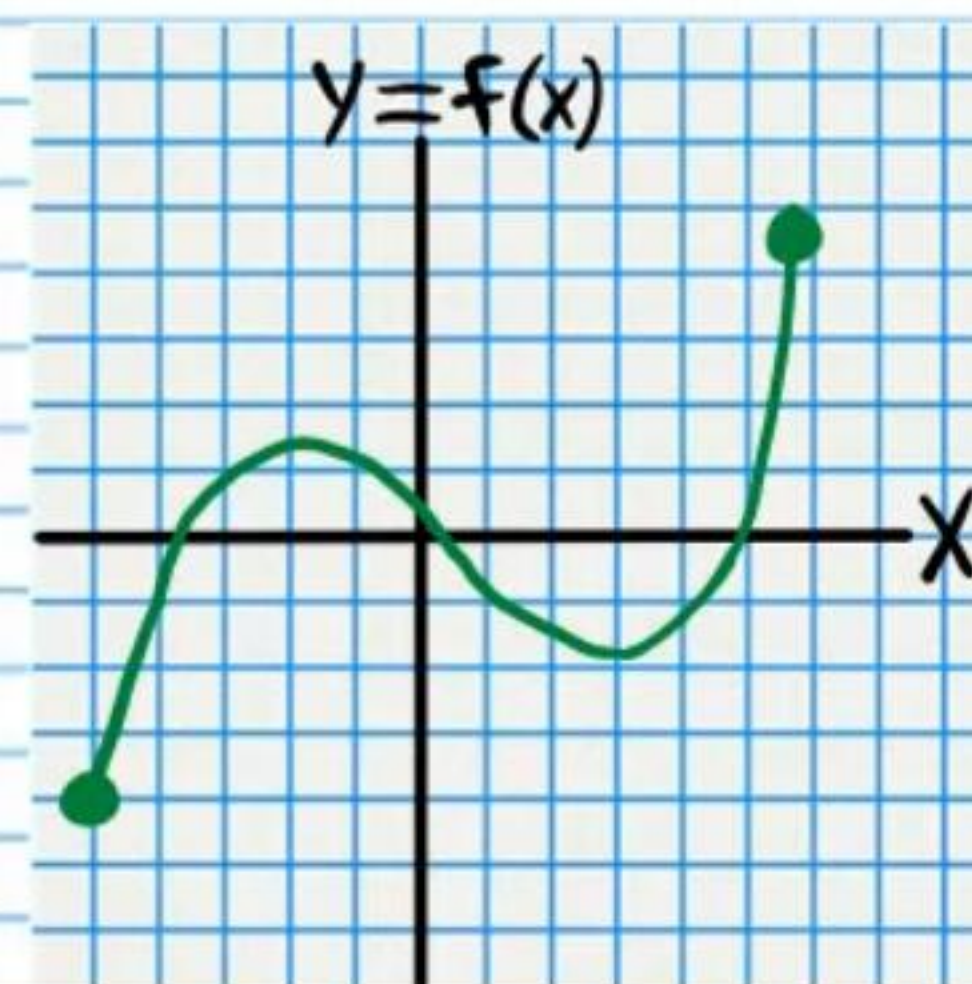
Values of a Function: The outputs of the function (i.e. the y-values of the function).

Closed Interval: An interval $[a, b]$ that includes all of the x-values between "a" and "b" as well as "a" and "b."

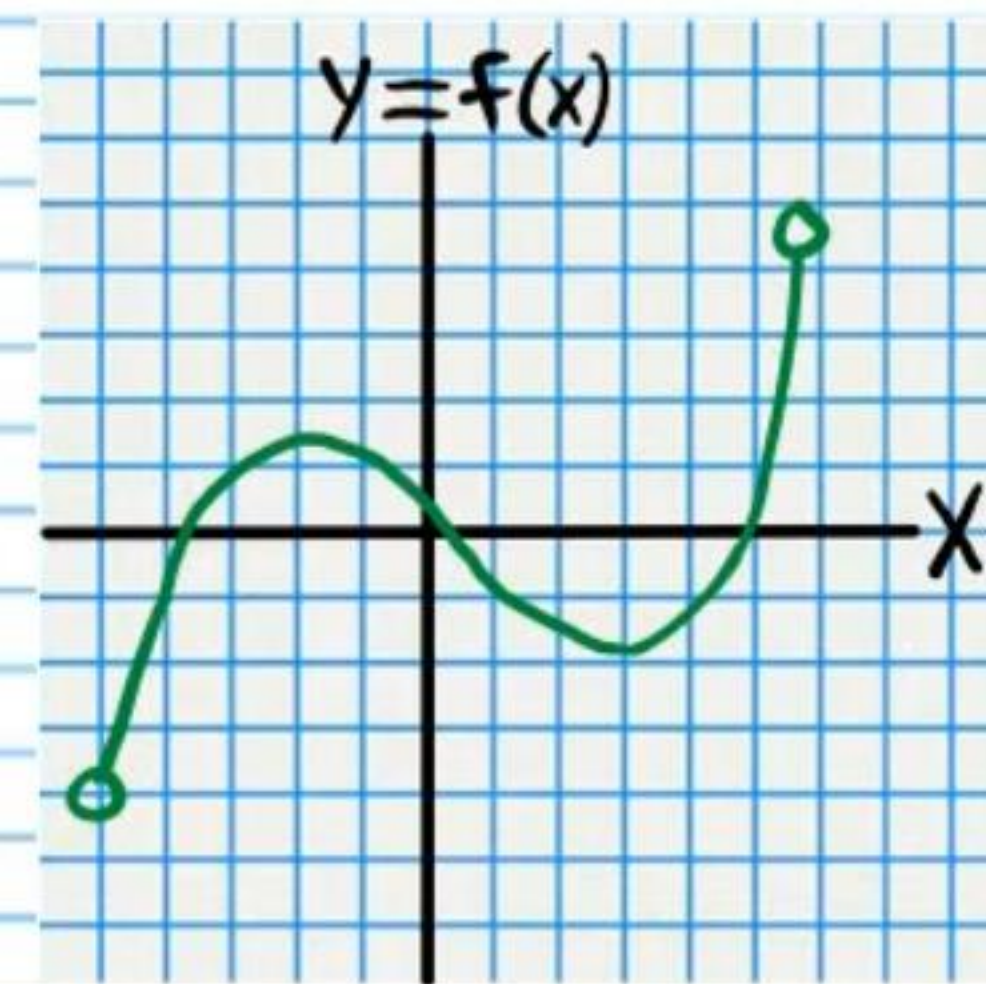
Open Interval: An interval (a, b) that includes all of the x-values between "a" and "b" but not "a" and "b."

Examples

Closed Interval



Open Interval

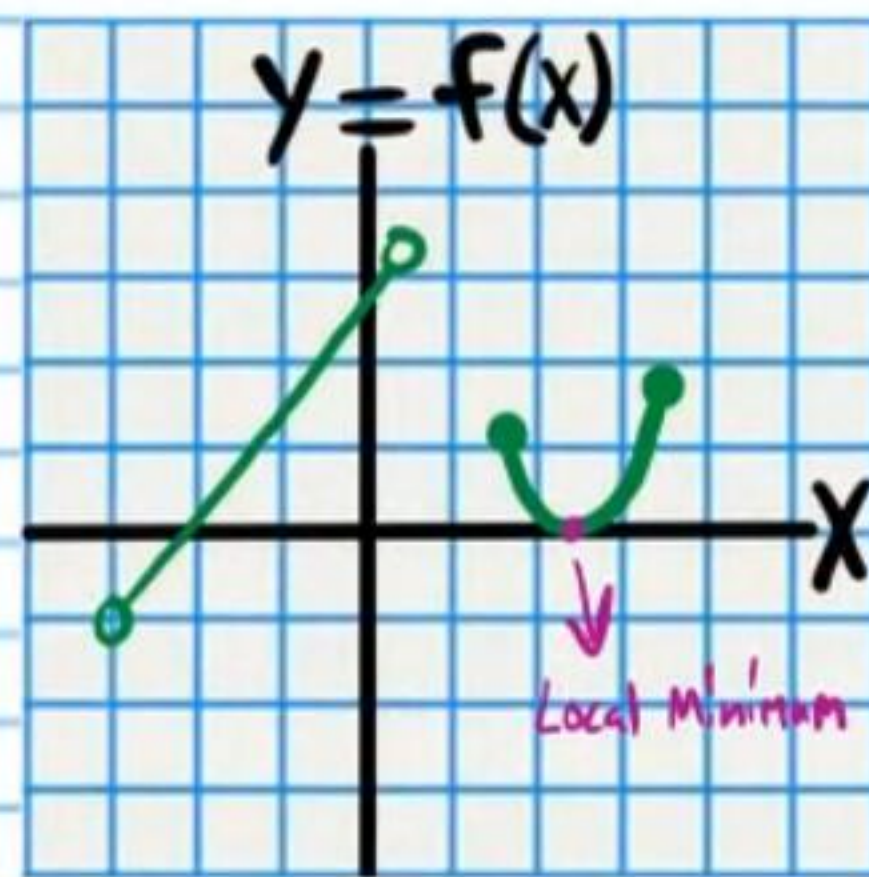
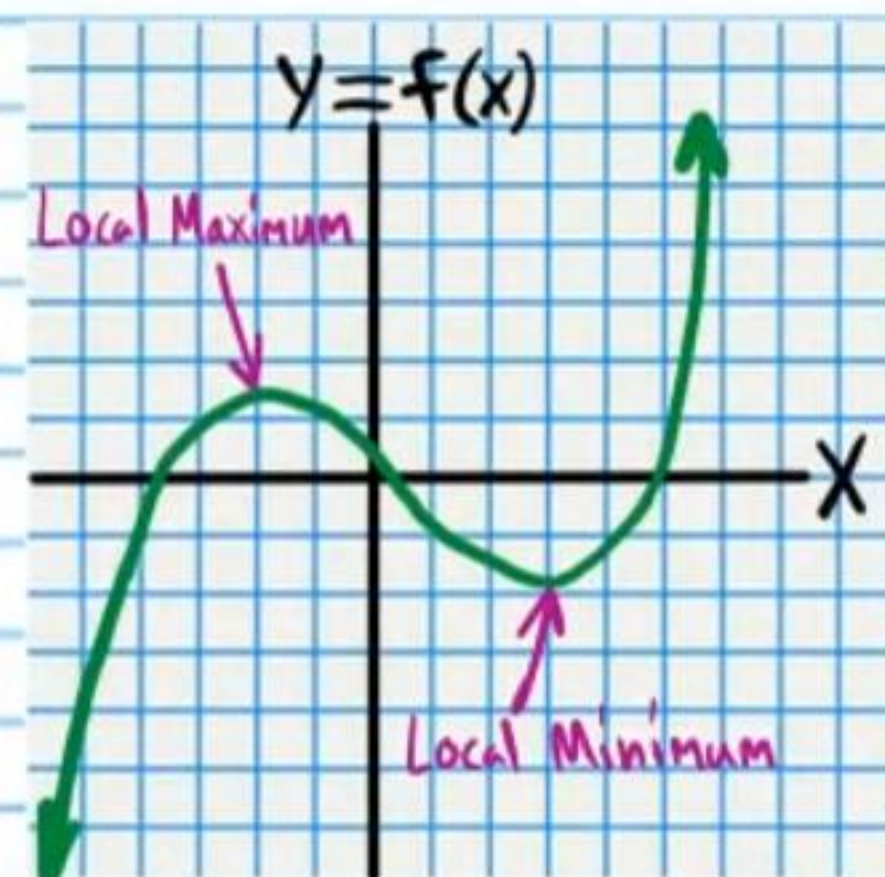


Extreme Value: An absolute maximum or an absolute minimum of a function (Sometimes refers to local maxima and minima too).

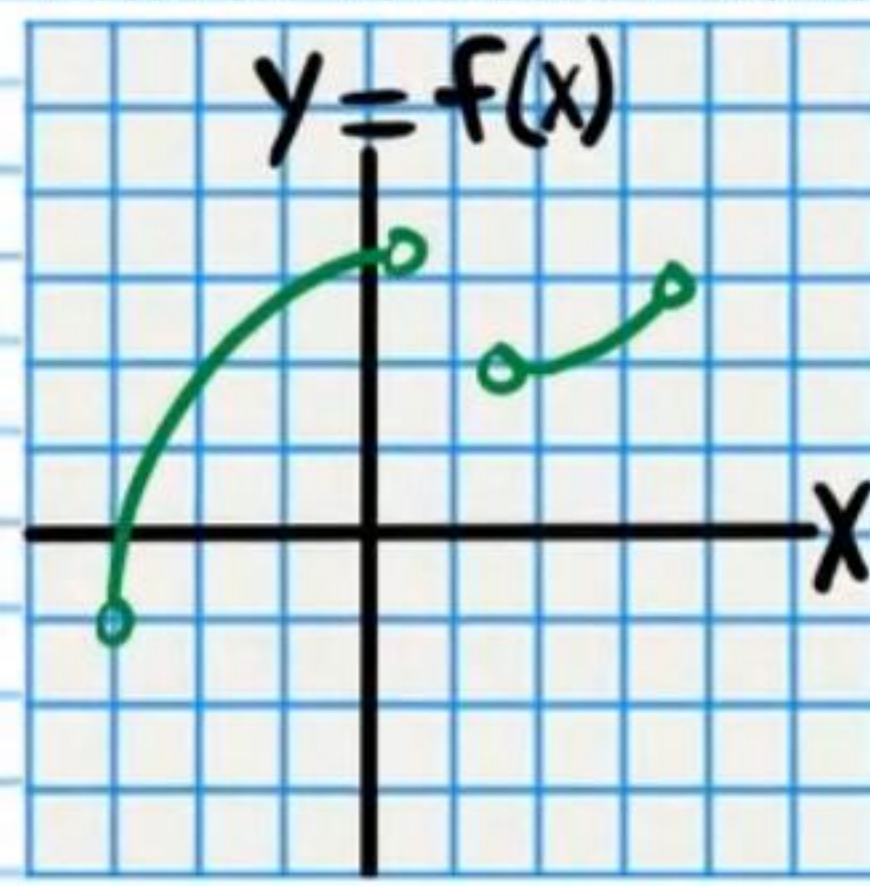
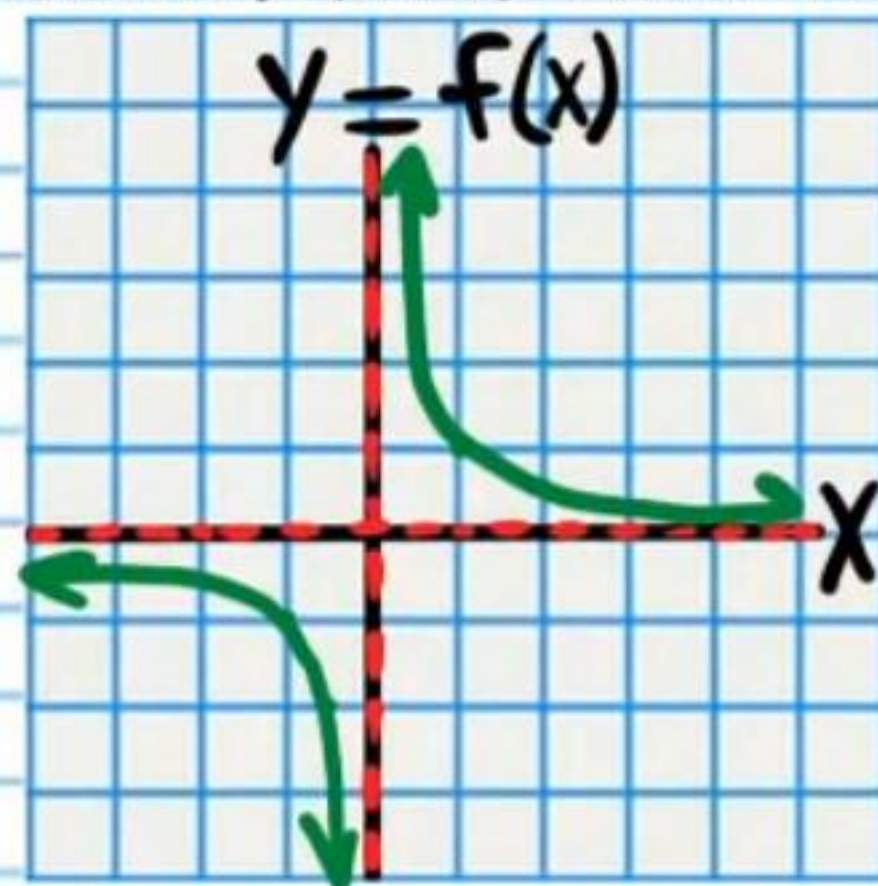
Extrema: Extreme values of a function.

The Extreme Value Theorem: If a function is continuous on a closed interval $[a, b]$, then the function has an absolute maximum and an absolute minimum at some x-values in $[a, b]$.

Situations where There Are No Absolute Maxima or Minima



Situations where There Are No Local or Absolute Maxima or Minima



Theorem for Finding the Absolute Maximum or Absolute Minimum of a Continuous Function on a Closed Interval

If a function is continuous on a closed interval, then the critical numbers and/or the endpoints are the locations where the absolute maximum value and the absolute minimum value of the function can be found.

Find the absolute maximum and the absolute minimum of each function on the specified interval.

★ A graphing calculator will help us with calculations and concepts.

● ⑬ $f(x) = x^4 + 4x^3 + 4x^2$, $[-4, 1]$

I) Finding Critical Numbers

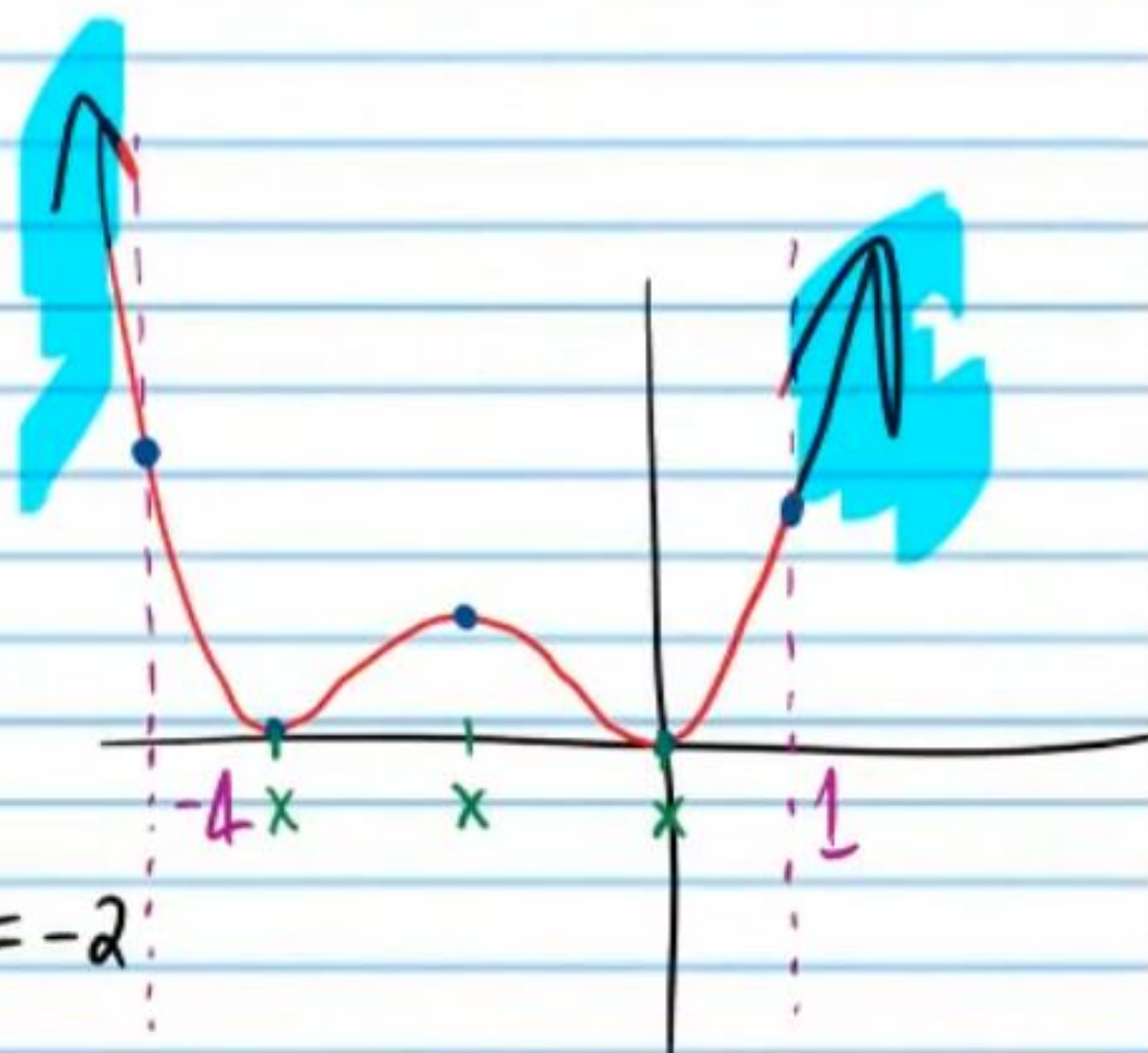
$$f'(x) = 4x^3 + 12x^2 + 8x$$

$$0 = 4x^3 + 12x^2 + 8x$$

$$0 = 4x(x^2 + 3x + 2)$$

$$0 = 4x(x+2)(x+1)$$

$x=0$ or $x=-1$ or $x=-2$



II) Evaluating the Function at the Critical Numbers

$$f(x) = x^4 + 4x^3 + 4x^2$$

$$f(0) = 0^4 + 4(0)^3 + 4(0)^2 = 0$$

$$f(-1) = (-1)^4 + 4(-1)^3 + 4(-1)^2 = 1 - 4 + 4 = 1$$

$$f(-2) = (-2)^4 + 4(-2)^3 + 4(-2)^2 = 0$$

III) Evaluating the Function at the Endpoints

$$f(x) = x^4 + 4x^3 + 4x^2$$

$$f(1) = 1^4 + 4(1)^3 + 4(1)^2 = 9$$

$$f(-4) = (-4)^4 + 4(-4)^3 + 4(-4)^2 = 64$$

Absolute Maximum: $\boxed{64}$ Absolute Minimum: $\boxed{0}$

⑭ $f(x) = \sqrt[3]{x^2} + \sqrt[3]{x^5} + 3$ $[-\frac{1}{5}, \frac{1}{10}]$

I) Finding Critical Numbers

$$f(x) = x^{\frac{2}{3}} + x^{\frac{5}{3}} + 3$$

$$f'(x) = \frac{2}{3}x^{-\frac{1}{3}} + \frac{5}{3}x^{\frac{2}{3}} + 0$$

$$f'(x) = \frac{2}{3x^{\frac{1}{3}}} + \frac{5}{3}\sqrt[3]{x^2}$$

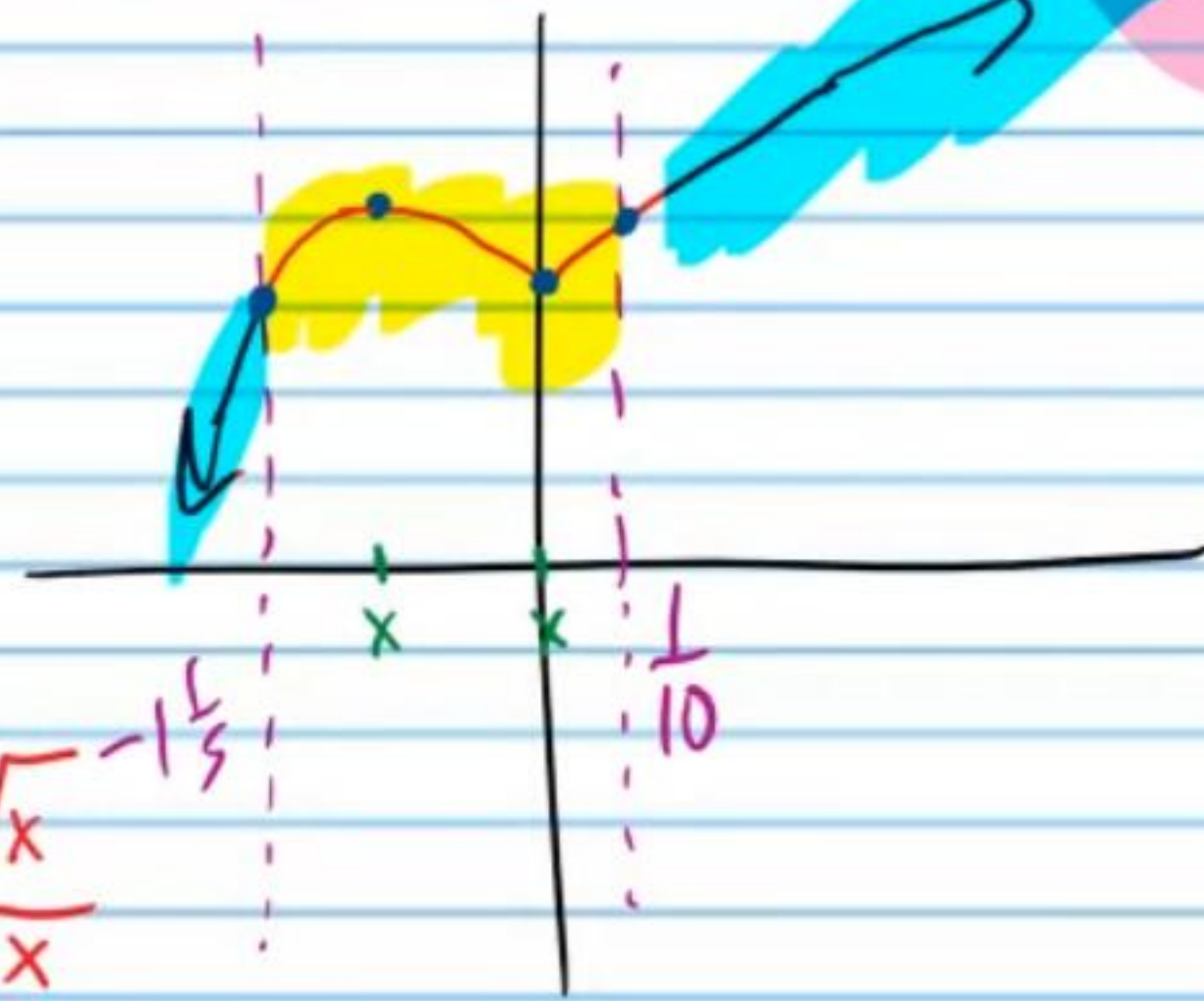
$$f'(x) = \frac{2}{3\sqrt[3]{x}} + \frac{5\sqrt[3]{x^2}}{3}$$

$$f'(x) = \frac{2}{3\sqrt[3]{x}} + \frac{5\sqrt[3]{x^2} \cdot \sqrt[3]{x}}{3 \cdot \sqrt[3]{x}}$$

$$f'(x) = \frac{2}{3\sqrt[3]{x}} + \frac{5\sqrt[3]{x^3}}{3\sqrt[3]{x}}$$

$$f'(x) = \frac{2}{3\sqrt[3]{x}} + \frac{5x}{3\sqrt[3]{x}}$$

$$f'(x) = \frac{2}{3\sqrt[3]{x}} + \frac{5x}{3\sqrt[3]{x}}$$



$$\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$$

$$f'(x) = \frac{2+5x}{3\sqrt[3]{x}}$$

$$\left. \begin{array}{l} 2+5x=0 \\ x=-\frac{2}{5} \end{array} \right\} \begin{array}{l} 3\sqrt[3]{x}=0 \\ x=0 \end{array}$$

II) Evaluating the Function at the Critical Numbers

$$f(x) = \sqrt[3]{x^2} + \sqrt[3]{x^5} + 3$$

$$f(0) = \sqrt[3]{0^2} + \sqrt[3]{0^5} + 3 = 3$$

$$f(-\frac{2}{5}) = \sqrt[3]{(-\frac{2}{5})^2} + \sqrt[3]{(-\frac{2}{5})^5} + 3 \approx 3.32573$$

III) Evaluating the Function at the Endpoints

$$f(x) = \sqrt[3]{x^2} + \sqrt[3]{x^5} + 3$$

$$f(-1.2) = \sqrt[3]{(-1.2)^2} + \sqrt[3]{(-1.2)^5} + 3 \approx 2.77415$$

$$f(0.1) = \sqrt[3]{(0.1)^2} + \sqrt[3]{(0.1)^5} + 3 \approx 3.23699$$

$$\text{Absolute Maximum: } 3.32573 \quad \text{Absolute Minimum: } 2.77415$$

15) $f(x) = \sqrt[7]{(2x+1)^5} + 1, [-2, 0]$

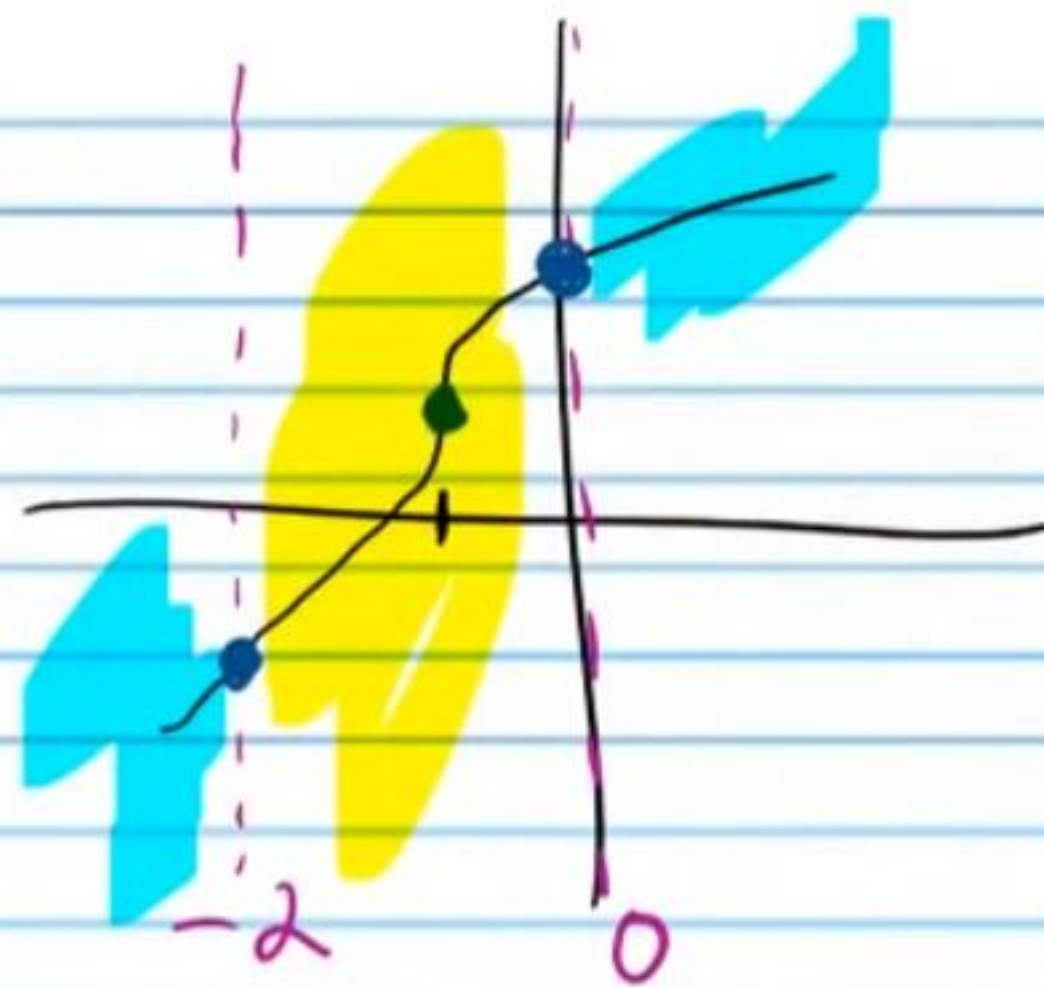
I) Finding Critical Numbers

$$f(x) = (2x+1)^{\frac{5}{7}} + 1$$

$$f'(x) = \frac{5}{7}(2x+1)^{-\frac{2}{7}}(2) + 0$$

$$f'(x) = \frac{5}{7(2x+1)^{\frac{2}{7}}}(2)$$

$$f'(x) = \frac{10}{7\sqrt[7]{(2x+1)^2}}$$



$$\sqrt[7]{(2x+1)^2} = 0$$

$$x = -\frac{1}{2}$$

II) Evaluating the Function at the Critical Numbers

$$f(x) = \sqrt[7]{(2x+1)^5} + 1$$

$$f(-\frac{1}{2}) = \sqrt[7]{(2(-\frac{1}{2})+1)^5} + 1 = 0 + 1 = 1$$

III) Evaluating the Function at the Endpoints

$$f(x) = \sqrt[7]{(2x+1)^5} + 1$$

$$f(0) = \sqrt[7]{(2(0)+1)^5} + 1 = 1 + 1 = 2$$

$$f(-2) = \sqrt[7]{(2(-2)+1)^5} + 1 \approx -1.1918$$

Absolute Maximum: $\boxed{2}$ Absolute Minimum: $\boxed{-1.1918}$

16) $f(x) = x^3 - 3x + 1, [-3, 1.5]$

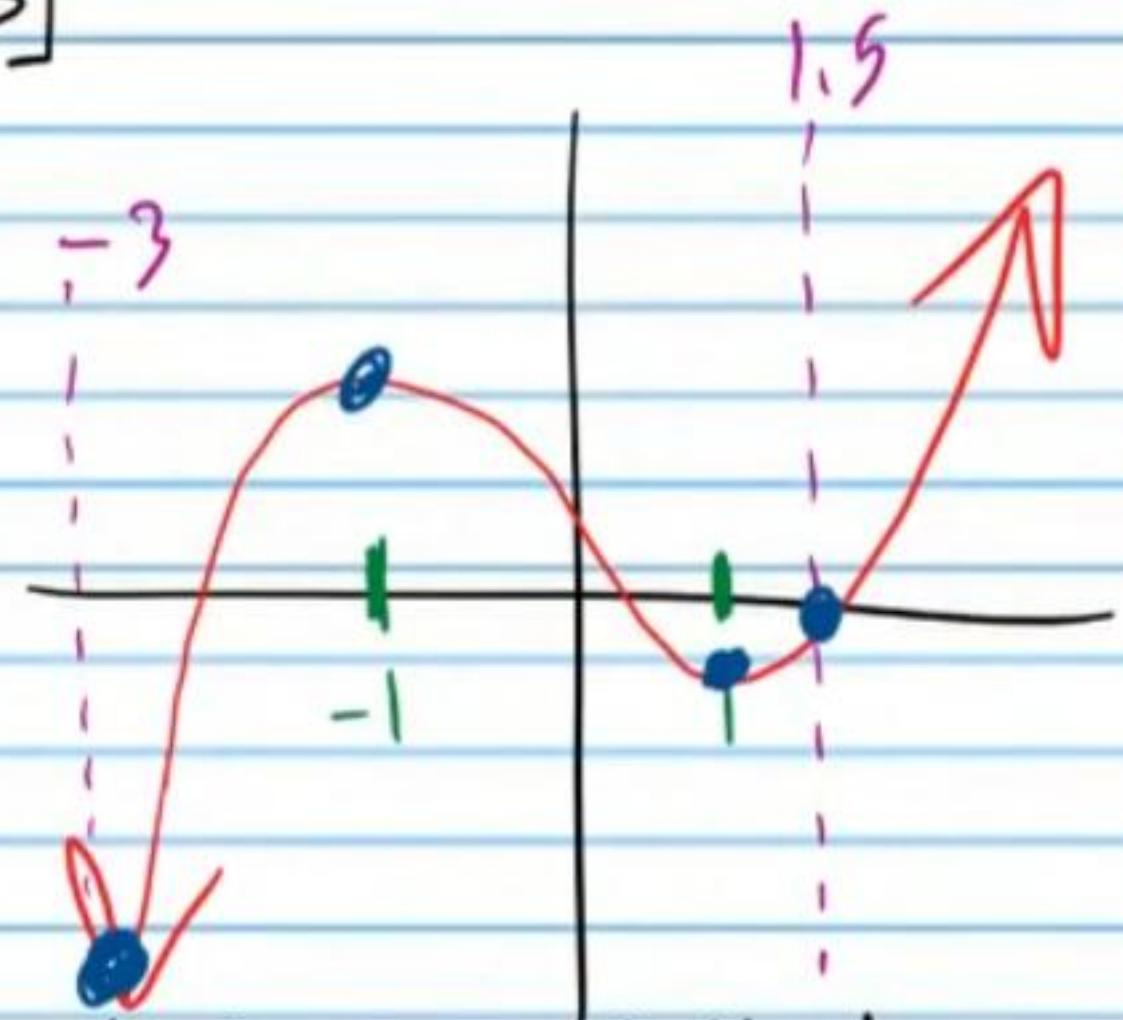
I) Finding Critical Numbers

$$f'(x) = 3x^2 - 3 + 0$$

$$0 = 3x^2 - 3$$

$$x^2 = 1$$

$$x = \pm 1$$



II) Evaluating the Function at the Critical Numbers

$$f(x) = x^3 - 3x + 1$$

$$f(1) = 1^3 - 3(1) + 1 = -1$$

$$f(-1) = (-1)^3 - 3(-1) + 1 = -1 + 3 + 1 = 3$$

III) Evaluating the Function at the Endpoints

$$f(x) = x^3 - 3x + 1$$

$$f(-3) = (-3)^3 - 3(-3) + 1 = -17$$

$$f(1.5) = (1.5)^3 - 3(1.5) + 1 = -0.125$$

Absolute Maximum: $\boxed{3}$ Absolute Minimum: $\boxed{-17}$

⑪ $f(x) = 2\sqrt[5]{x^4} - 8$ $[-1, 2]$

I) Finding Critical Numbers

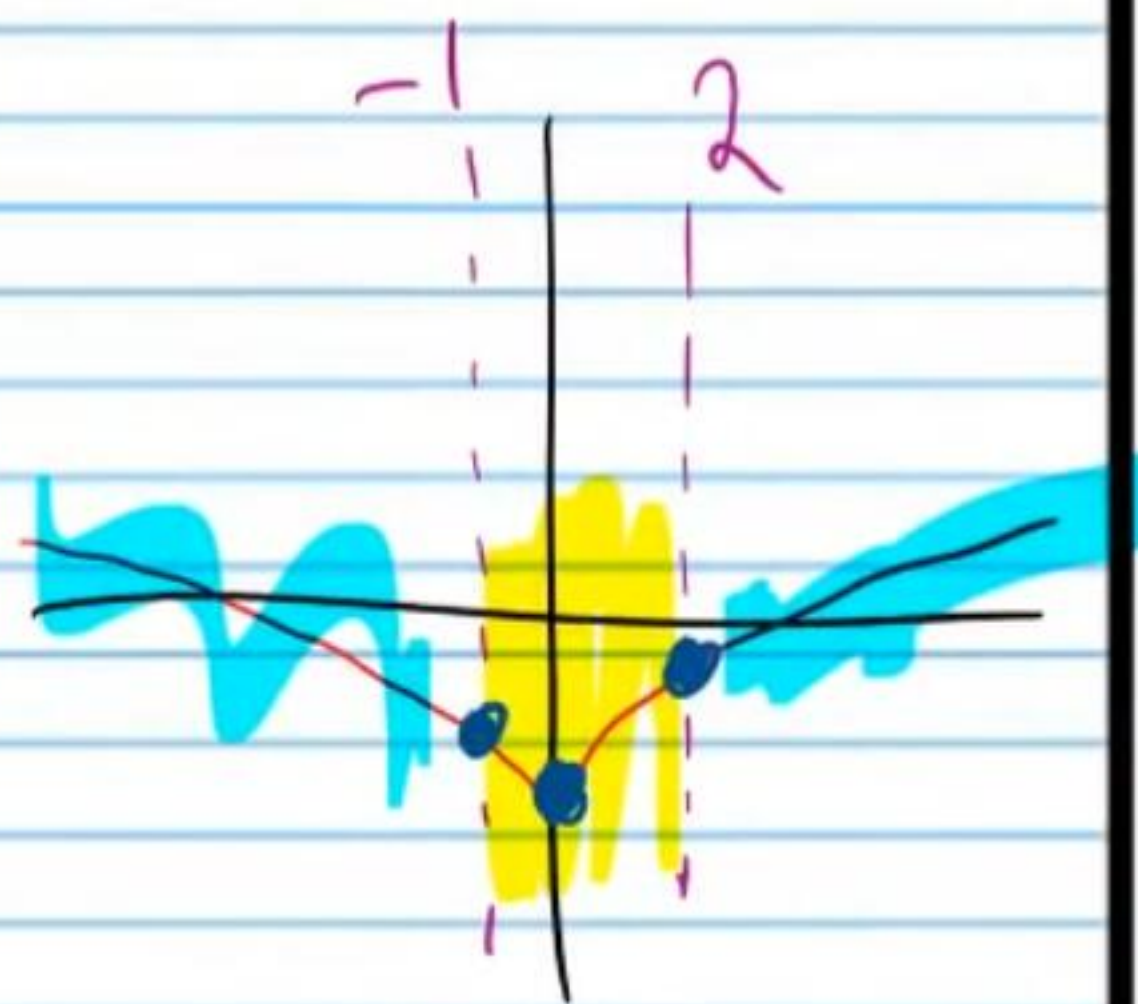
$$f(x) = 2x^{4/5} - 8$$

$$f'(x) = \frac{8}{5}x^{-1/5} - 0$$

$$f'(x) = \frac{8}{5\sqrt[5]{x}}$$

$$5\sqrt[5]{x} = 0$$

$$x = 0$$



II) Evaluating the Function at the Critical Numbers

$$f(x) = 2\sqrt[5]{x^4} - 8$$

$$f(0) = 2\sqrt[5]{0^4} - 8 = 0 - 8 = -8$$

III) Evaluating the Function at the Endpoints

$$f(x) = 2\sqrt[5]{x^4} - 8$$

$$f(-1) = 2\sqrt[5]{(-1)^4} - 8 = 2 - 8 = -6$$

$$f(2) = 2\sqrt[5]{2^4} - 8 = -4.5178$$

Absolute Maximum: -4.5178 Absolute Minimum: -8

★ You must use the chain rule when finding the derivative.

18) $f(x) = x - 3 - 10\sqrt[3]{\frac{4}{5}x - 3}$, $[-4, 10]$ ★ Hint: Make your horizontal window on your calculator $x_{\min}: -15$ $x_{\max}: 15$

I) Finding Critical Numbers

$$f(x) = x - 3 - 10\left(\frac{4}{5}x - 3\right)^{\frac{1}{3}}$$

$$f'(x) = 1 - 0 - \frac{10}{3}\left(\frac{4}{5}x - 3\right)^{-\frac{2}{3}}\left(\frac{4}{5}\right)$$

$$f'(x) = 1 - \frac{10}{3\left(\frac{4}{5}x - 3\right)^{\frac{2}{3}}}\left(\frac{4}{5}\right)$$

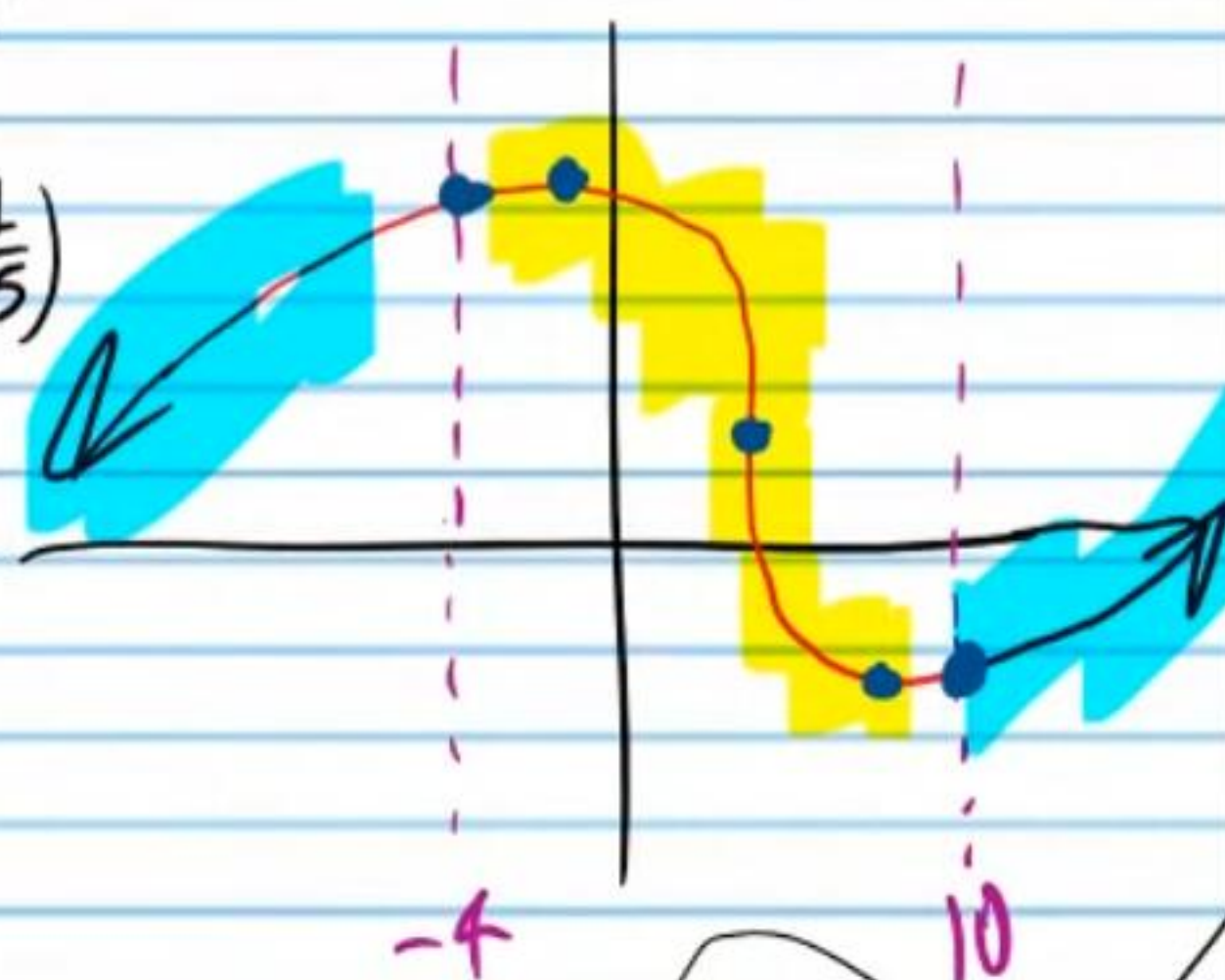
$$f'(x) = 1 - \frac{40}{15\sqrt[3]{\left(\frac{4}{5}x - 3\right)^2}}$$

$$f'(x) = 1 - \frac{8}{3\sqrt[3]{\left(\frac{4}{5}x - 3\right)^2}}$$

$$f'(x) = \frac{1 \cdot 3\sqrt[3]{\left(\frac{4}{5}x - 3\right)^2}}{3\sqrt[3]{\left(\frac{4}{5}x - 3\right)^2}} - \frac{8}{3\sqrt[3]{\left(\frac{4}{5}x - 3\right)^2}}$$

$$f'(x) = \frac{3\sqrt[3]{\left(\frac{4}{5}x - 3\right)^2}}{3\sqrt[3]{\left(\frac{4}{5}x - 3\right)^2}} - \frac{8}{3\sqrt[3]{\left(\frac{4}{5}x - 3\right)^2}}$$

$$f'(x) = \frac{3\sqrt[3]{\left(\frac{4}{5}x - 3\right)^2} - 8}{3\sqrt[3]{\left(\frac{4}{5}x - 3\right)^2}}$$



$$3\sqrt[3]{\left(\frac{4}{5}x - 3\right)^2} = 0$$

$$\sqrt[3]{\left(\frac{4}{5}x - 3\right)^2} = 0$$

$$\left(\frac{4}{5}x - 3\right)^2 = 0$$

$$\frac{4}{5}x - 3 = \pm 0$$

$$x = \frac{15}{4}$$

$$3\sqrt[3]{\left(\frac{4}{5}x - 3\right)^2} - 8 = 0$$

$$\sqrt[3]{\left(\frac{4}{5}x - 3\right)^2} = \frac{8}{3}$$

$$\left(\frac{4}{5}x - 3\right)^2 = \frac{512}{27}$$

$$\frac{4}{5}x - 3 = \pm \sqrt{\frac{512}{27}}$$

$$\frac{4}{5}x - 3 \approx \pm 4.3546$$

$$x \approx 9.1933 \text{ or } x \approx -1.6933$$

II) Evaluating the Function at the Critical Numbers

$$f(x) = x - 3 - 10\sqrt[3]{x - 3}$$

$$f(9.1933) \approx -10.1366$$

$$f(-1.6933) \approx 11.6366$$

$$f\left(\frac{15}{4}\right) \approx 0.75$$

III) Evaluating the Function at the Endpoints

$$f(x) = x - 3 - 10\sqrt[3]{x - 3}$$

$$f(-4) \approx 11.37$$

$$f(10) \approx -10.09976$$

$$\text{Absolute Maximum: } 11.6366 \quad \text{Absolute Minimum: } -10.1366$$